AquaPearl 500 Pro — Product Manual

Document ID: MAN-AP500-2026.1 | Version 2026.1 | January 2026

Pearlz Home Systems (Pvt.) Ltd. This manual covers box contents, specifications, controls, operation, filter intervals, cleaning, safety, and audible alert codes E-01 to E-08.

1. Box contents

- AquaPearl 500 Pro main unit
- Sediment pre-filter cartridge (pre-installed)
- Carbon filter cartridge
- RO membrane cartridge
- UV lamp module (factory installed)
- Inlet and outlet tubing kit (2 m)
- Adaptor valves and mounting screws
- Power adapter 24V DC
- Quick-start card and warranty leaflet

2. Technical specifications

Parameter	Value
Flow rate (purified)	Up to 15 L/hour
Tank capacity	8 litres storage
Dimensions (W×D×H)	40 × 25 × 48 cm
Power	24V DC, 65W peak
Inlet pressure	0.1 – 0.4 MPa
Inlet water temp	5 – 38 °C
Filter stages	Sediment, Carbon, RO, UV
Max inlet TDS	1500 ppm

3. Control panel and indicator lights

Power LED (white): solid = on. Status LED: green = normal; amber = maintenance due; blinking red = fault (see alert codes). TDS display shows purified water estimate in ppm. RESET button clears certain alerts after corrective action. PAIR button enables Bluetooth pairing.

4. Operating instructions

1. Ensure inlet tap is open and power is connected. 2. Wait for the tank to fill on first run. 3. Dispense from the purified water tap. 4. Check the app dashboard weekly for filter life. 5. Do not store hot liquids in the tank. 6. Keep 10 cm clearance around vents.

5. Filter replacement intervals

Filter	Typical interval	SKU
Sediment pre-filter	3–6 months or on E-03	FLT-SED-01
Carbon filter	6–12 months	FLT-CAR-01
RO membrane	18–24 months	FLT-RO-01
UV lamp	24 months or on E-08	FLT-UV-01

6. Cleaning and descaling schedule

Wipe the exterior weekly with a damp cloth. Every 3 months, run a descaling cycle using Pearlz Descale Solution (or food-grade citric acid per quick-start card): fill the tank with solution, run a complete dispense-and-refill cycle twice, then flush with two full tanks of clean water. Never use abrasive cleaners or bleach on internal wet paths.

7. Safety warnings

- Do not immerse the main unit in water.
- Use only the supplied 24V adapter.
- Disconnect power before filter changes.
- Keep away from children during filter replacement.
- If you smell burning or see smoke, unplug immediately and contact support.

8. Audible alert codes (E-01 to E-08)

Filter maintenance section reference. Three short beeps indicate E-03 sediment pre-filter saturation. Replace the pre-filter cartridge, then press and hold RESET for five seconds.

Code	Pattern	Meaning
E-01	2 long beeps	Inlet valve timeout
E-02	1 long + 1 short	Tank full sensor fault
E-03	3 short beeps	Sediment pre-filter saturation
E-04	4 short beeps	Low inlet pressure
E-05	Blinking red LED	Leak sensor wet
E-06	Continuous 2s tone	Over-temperature (firmware 2.4+)
E-07	Alternating beeps	Communication fault
E-08	5 rapid beeps	UV lamp failure — escalate

Filtration performance: multi-stage RO+UV reduces dissolved solids and addresses microbiological risk under specified inlet conditions. Always follow local boil-water advisories when issued by authorities. This manual does not claim any third-party certification mark.

9. Daily use tips

Dispense water only from the purified tap. Avoid leaving the inlet closed for more than seven days without a flush cycle. If the unit will be unused for over two weeks, drain the tank and power off. On return, run a full flush before drinking.

Do not store ice cubes made from unflushed water after a filter change. Taste issues after cartridge replacement are almost always resolved by completing the Installation Guide first-run flush (discard the first two full tanks).

Keep serial numbers photographed with the invoice. The rating label is on the rear panel above the power inlet. Support requests without a serial number take longer to process.

10. Storage tank care

The 8-litre storage tank is food-contact safe plastic. Empty and wipe the tank interior every six months. Do not store flavoured beverages in the tank.

If the tank develops biofilm odour, perform a citric-acid descaling cycle followed by two full flush tanks. Never use chlorine bleach inside the purified water path.

Tank capacity is nominal; usable volume is slightly less when the float sensor cuts inlet fill. Continuous pump running with a full tank may indicate a stuck float — see Troubleshooting.

11. App connectivity overview

The Pearlz app shows filter life, TDS, and firmware version. Pairing requires Bluetooth. Firmware updates install overnight when the unit is idle and powered.

Hold PAIR for three seconds until the status LED blinks blue. Keep the phone within two metres during pairing. If pairing fails, reboot the phone Bluetooth stack and retry once.

App filter-life percentages are estimates based on flow and time. An E-03 alert overrides the estimate: replace the sediment pre-filter even if the app still shows remaining life.

12. Spare parts philosophy

Use only Pearlz SKUs listed in the Pricing Guide. Non-Pearlz cartridges may void warranty for damage attributed to those parts.

Sediment FLT-SED-01, carbon FLT-CAR-01, RO FLT-RO-01, and UV FLT-UV-01 are the supported consumables for AquaPearl 500 Pro. Intervals are in Section 5; prices are not listed here.

UV lamp replacement is recommended via an Authorised Service Partner. Customer self-install of UV modules can leave the chamber misaligned and will void related coverage.

13. Water quality assumptions

The unit expects cold municipal or well water within the inlet pressure and TDS limits in Section 2. It is not designed for seawater, untreated industrial effluent, or hot-water lines.

High inlet TDS shortens RO membrane life. If purified TDS rises sharply, check membrane age and inlet TDS before assuming a sensor fault.

This manual discusses filtration performance under specified conditions. It does not claim any third-party certification mark such as NSF. Do not invent certification language.

14. Seasonal maintenance calendar

Spring: inspect tubing for UV damage if the unit is near a window; wipe vents.

Monsoon: check inlet strainer more often; sediment load rises and E-03 may trigger earlier.

Winter: if inlet temperature approaches 5 °C, expect slower production; do not add heaters inside the cabinet without Pearlz guidance.

Annually: consider AMC Plus or Premium if you prefer scheduled Service Partner visits (see Service & Maintenance Agreement).

15. Relocation and moving house

Before moving: close inlet, drain tank, disconnect power, cap open tubing ends.

After relocating: leak-check all joints, run a full first-run flush, and re-pair the app.

Warranty remains with the unit when transfer rules in the Warranty Policy are followed. Extended warranties on discontinued models are not transferable.

16. Indicator light quick reference

White power LED solid: unit powered. Off: no power or deep sleep after extended idle.

Green status: normal operation. Amber: maintenance due soon. Blinking red: active fault (map pattern to E-codes in Section 8).

Blue blink on status during PAIR: Bluetooth advertising mode active for two minutes.

17. Descaling procedure detail

Every three months under hard-water conditions, run Pearlz Descale Solution or food-grade citric acid per the quick-start card. Fill the tank with solution, complete two dispense-and-refill cycles, then flush with two full tanks of clean water.

Never mix descaler with bleach. Never pour boiling water into the tank. If white scale remains on the dispense spout exterior, wipe with a soft cloth only.

18. End-of-life recycling

Return end-of-life units through Pearlz take-back weeks announced on the website, or dispose electronics via certified e-waste channels in your city.

Remove batteries from any remote accessories before recycling. Filters may be bagged separately according to local solid-waste rules.

19. Glossary

TDS: total dissolved solids, estimated in ppm on the display for AquaPearl 500 Pro.

RO: reverse osmosis membrane stage. UV: ultraviolet lamp stage on 500 Pro only.

E-03: three short beeps indicating sediment pre-filter saturation — replace FLT-SED-01 then hold RESET five seconds.

20. Document control

This manual version 2026.1 supersedes 2025.4. Alert code E-06 meaning follows firmware 2.4+ as documented in the Product Changelog. Always confirm firmware in the app before advising on E-06.

For warranty, refund, and shipping terms, use the dedicated policy documents — this manual does not redefine commercial policy windows.